

THEJASEKHAR REDDY GUNDLOORU

Senior Site Reliability Engineer | Cloud Platform & Observability Engineering

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PROFESSIONAL SUMMARY

Platform and observability engineer with 3 years building and operating production-grade telemetry and infrastructure platforms for enterprise banking clients, including large-scale OpenTelemetry pipelines processing 2M+ traces/min and 50M+ metrics/min on Kubernetes. Strong hands-on background in container orchestration, CI/CD automation, and observability stack engineering (OpenTelemetry, Elasticsearch, ClickHouse, Apache Doris, Kafka) on AWS. Builder of multiple open-source platform tools spanning fleet management, log analytics, and SRE tooling. Currently deepening expertise in Terraform/IaC, GitOps, and Internal Developer Platforms to extend platform engineering impact at scale.

CORE TECHNICAL SKILLS

- Cloud & Container Platforms: AWS (EC2, IAM, VPC, S3), Kubernetes (cluster operations, networking, RBAC, troubleshooting, scaling), GCP (GKE), Docker, Linux Administration (RHEL/Ubuntu)
- Observability & Telemetry: OpenTelemetry (Collector, OpAMP, instrumentation), Elasticsearch, ClickHouse, Grafana, Apache Doris, Prometheus-style metrics pipelines, Kafka
- CI/CD & Automation: GitHub Actions, GitLab CI, Python & Bash scripting
- Reliability Engineering: SLO/SLI design, burn-rate alerting, RCA workflows, MTTR reduction, ML-based anomaly detection
- Pipelines: Kafka, Logstash, Fluent Bit, Filebeat, Metricbeat — ETL/ELT, adaptive sampling, high-volume log processing.
- Platform Engineering (in progress): Terraform/IaC, GitOps (ArgoCD/Flux), Backstage (IDP), Istio, policy-as-code (Kyverno)
- Languages & Frameworks: Python, Go, JavaScript, SQL, Shell/Bash scripting.

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Platform Engineering & Observability | HCLSoftware | Bengaluru, India *4 Mar 2024 – Present*

- Engineered OpenTelemetry-based observability pipelines for enterprise banking clients (including Mashreq Bank), sustaining 2M+ traces/min and 50M+ metrics/min while maintaining 99.95% platform uptime.
- Acted as the primary owner for the Kubernetes platform infrastructure lifecycle, implementing comprehensive in-place observability to ensure continuous cluster health, rapid troubleshooting, and operational excellence.
- Operated and scaled Kubernetes infrastructure underpinning telemetry collection and processing services, handling cluster troubleshooting, resource management, and workload reliability for production banking workloads.
- Built log analytics pipelines processing 500GB+/day through Elasticsearch and ClickHouse, supporting compliance and security monitoring requirements.
- Reduced root-cause analysis time by 40% and mean-time-to-resolution by 35% by correlating logs, metrics, and traces across the OpenTelemetry pipeline.
- Managed the end-to-end lifecycle of enterprise-scale Elasticsearch clusters for Mashreq, executing zero-downtime upgrades and continuous patching to ensure platform resilience.
- Implemented Cross-Cluster Replication (CCR) for disaster recovery and Cross-Cluster Search (CCS) to enable centralized observability across distributed environments.
- Deployed an ML-based anomaly detection model achieving 78% precision for proactive incident identification, reducing reliance on manual threshold alerting.
- Automated observability infrastructure deployment sequences, strictly prioritizing component rollout—ClickHouse, followed by Kafka, OpenTelemetry Gateway (OTel-GW), and Grafana—ensuring telemetry pipeline stability and resilience.
- Integrated automated CI/CD pipelines to build, package, and deploy custom Grafana plugins and dashboards-as-code from VCS directly into development and production environments.

- Developed automation scripts to provision high-availability, on-premise Kubernetes clusters, streamlining the configuration of multi-node control and worker planes.
- Developed and standardized reusable Terraform modules for AWS infrastructure, powering the backend provisioning engine of a self-service Internal Developer Platform (IDP).
- Designed and maintained CI/CD pipelines using GitHub Actions implementing multi-environment, blue-green, and canary deployment strategies to improve release reliability and delivery velocity.
- Developed an AI-driven observability assistant (POC) by integrating LibreChat with a ClickHouse Model Context Protocol (MCP) server connected to production APM databases. This setup allowed engineering teams to query complex telemetry datasets using natural language, automated the generation of optimized ClickHouse SQL queries, and streamlined contextual troubleshooting.
- Mentored team members on OpenTelemetry instrumentation and observability best practices, and contributed to platform engineering standards across the team.

Key Projects

Enterprise Log Analytics & Predictive Monitoring — Mashreq Bank

Stack: Elasticsearch, Logstash, Kibana, Beats, Elastic ML, Kafka

- Engineered Logstash/Beats ingestion workflows normalising 500GB+/day from 200+ banking data sources — improving log queryability by 95%.
- Applied Elastic ML to detect capacity anomalies proactively; prevented 12 customer-impacting incidents in 6 months with 78% precision on critical service signals.
- Introduced Kafka-based decoupling to handle burst traffic, improving ingestion resilience and SLA compliance to 99.95%.
- Implemented Elasticsearch Cross-Cluster Replication (CCR) between primary and DR clusters for disaster recovery, and Cross-Cluster Search (CCS) to enable unified querying across both clusters without duplicating sensitive banking data.
- Designed structured log schemas and enrichment pipelines enabling low-latency analytics and reducing forensic investigation time from hours to seconds.

Enterprise Kubernetes Observability Platform

Stack: Kubernetes, OTEL Operator & Gateway, Elasticsearch, Kibana, Prometheus, Grafana

- Deployed OTEL Operator across 15+ clusters with automatic sidecar and daemonset injection — achieving 95%+ coverage without code changes.
- Centralised OTEL Gateway pipelines processing 2M+ traces/min, 50M+ metrics/min, and 500GB+/day logs at 99.9% delivery SLA.
- Defined SLOs, SLIs, and error budgets using golden signals (latency, traffic, errors, saturation) to drive reliability decisions.

Highly Available Kubernetes Platform

Stack: Ubuntu 22.04, Kubeadm, etcd, HAProxy, Keepalived, Prometheus, HPA/VPA

- Provisioned a production-grade HA cluster — 3-node etcd control plane with HAProxy + Keepalived VIP failover — eliminating all single points of failure.
- Achieved 99.95% uptime across 50+ workloads reduced API server response latency by 25%.
- Integrated Prometheus + Alertmanager + HPA/VPA for proactive capacity management and failure prediction.

HCLMyCloud — Self-Service Cloud Infrastructure Platform

Stack: Terraform, AWS, .NET, Internal Developer Platform (IDP)

- Engineered a self-service Internal Developer Platform featuring a drag-and-drop UI and frontend APIs built on .NET, empowering engineering teams to dynamically provision cloud infrastructure and driving a "zero-ticket" operational model.
- Developed and standardized reusable Infrastructure as Code (IaC) modules using Terraform to power the backend provisioning engine of a .NET-based, drag-and-drop Internal Developer Platform.
- Built highly scalable and secure Terraform modules enabling automated, self-service provisioning for AWS infrastructure, ensuring cloud governance and standardizing resource deployments.
- Contributed to a "zero-ticket" operational model by enabling engineering teams to dynamically provision AWS infrastructure without manual operational overhead.

[R&D] AI-Driven Observability Assistant

Stack: LibreChart, ClickHouse MCP, ClickHouse, APM Telemetry, AI/LLM, Kubernetes

- Containerized and orchestrated the LibreChat and ClickHouse MCP stack on Kubernetes.
- Built an AI assistant connecting directly to production APM telemetry via ClickHouse MCP.
- Enabled engineers to query complex APM data using natural language, auto-generating optimized ClickHouse SQL to accelerate troubleshooting.

[R&D] Dynamic Elasticsearch Cluster Provisioning

Stack: Docker, Docker Compose, Bash

- Automated on-demand cluster provisioning, cutting setup time from 4 hours to 15 minutes and enabling self-service for 6 platform teams.
- Introduced configuration templating to enforce environment parity across dev, test, and staging, reducing human error.

OPEN-SOURCE & INDEPENDENT PROJECTS

Argus — *Observability fleet management for OpenTelemetry agents (Go, React, ClickHouse, OpAMP)*

- Built a fleet management platform for OpenTelemetry agents using the OpAMP protocol for bi-directional communication, enabling health monitoring, remote configuration management, and centralized pipeline visualization.
- Eliminated configuration drift and manual update processes by centralizing config management across the observability fleet, and reduced operational toil by surfacing previously silent agent failures.

LogScope — *Full-stack log analysis platform (Apache Doris, OTel Collector, FastAPI, React/Vite)*

- Built a log analysis platform on Apache Doris 4.x using the OTel-native schema, with an OTel Collector ingesting logs via Doris Stream Load.
- Implemented automated log pattern detection using Drain3 within a FastAPI backend, paired with a React/Vite frontend; resolved a Node 20+/ajv dependency collision by migrating the build to Vite 5.x on Node 18-slim.

CERTIFICATIONS & TRAINING

- ClickHouse Certified Developer | ClickHouse Observability Associate | ClickHouse Observability Professional
- AWS Certified Developer – Associate | AWS Academy Cloud Foundations
- Elastic Observability Engineer Training | Elasticsearch Engineer Training
- Confluent Data Streaming Engineer
- Introduction to OpenTelemetry (Linux Foundation) | Introduction to Istio (Linux Foundation)
- Kubernetes Certified Application Developer (KodeKloud) | OpenTelemetry Certified Associate (KodeKloud) | Golang Developer (KodeKloud)

COMMUNITY & CONTENT

- Author of a free practitioner-focused observability learning resource (thejareddy1.github.io/learn-observability) covering OpenTelemetry, Kubernetes, and observability pipeline architecture.

EDUCATION

B.Tech in Computer Science and Engineering — CGPA: 8.6